Leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One split-and-information-boundary record.

Trace assignments, repair overlap and explain the claim the split can support.

Keep one habit: W1–W3/W5 → selected plant split → project boundary record; scaling remains an optional named extension.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

Sampling and Intended Use

> **My remembered idea, example and correction**

Inspect Before Modeling

> **My remembered idea, example and correction**

Features, Labels, Targets

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A–F fixture: A1 A2 A3/B1 B2/C1 train;D1 D2 validation;E1 E2/F1 F2 test. Frames6/2/4 and groups3/1/2; code prints[] then3 1 2. D2 moved to training gives frames7/1/4 and shared D. For new routes North(A,B)/South(C,D), hold entire routes out rather than mixing route identity; two routes remain weak evidence.

  Separate pictured Cedar/Elm/Fir have3views each: train6views/2plants,test3/1; moving Fir view1 into training compromises unseen-plant evaluation.

- Explain what **training**, validation and **test set**s are used for. What changes if you repeatedly choose a model using its test score?

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A–F fixture: A1 A2 A3/B1 B2/C1 train;D1 D2 validation;E1 E2/F1 F2 test. Frames6/2/4 and groups3/1/2; code prints[] then3 1 2. D2 moved to training gives frames7/1/4 and shared D. For new routes North(A,B)/South(C,D), hold entire routes out rather than mixing route identity; two routes remain weak evidence.

  Separate pictured Cedar/Elm/Fir have3views each: train6views/2plants,test3/1; moving Fir view1 into training compromises unseen-plant evaluation.

- Reference for this exercise: open this week's student lesson, "Leakage", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked A through F plan, give **training**, validation and test counts first in **group**s and then in rows. Interpret [] from the code and state one problem it cannot detect.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A–F fixture: A1 A2 A3/B1 B2/C1 train;D1 D2 validation;E1 E2/F1 F2 test. Frames6/2/4 and groups3/1/2; code prints[] then3 1 2. D2 moved to training gives frames7/1/4 and shared D. For new routes North(A,B)/South(C,D), hold entire routes out rather than mixing route identity; two routes remain weak evidence.

  Separate pictured Cedar/Elm/Fir have3views each: train6views/2plants,test3/1; moving Fir view1 into training compromises unseen-plant evaluation.

- A1 and A2 are in **training** and A3 is in test. Explain why unique frame IDs and a fixed random seed do not fix this. Propose a repair and what to record.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A–F fixture: A1 A2 A3/B1 B2/C1 train;D1 D2 validation;E1 E2/F1 F2 test. Frames6/2/4 and groups3/1/2; code prints[] then3 1 2. D2 moved to training gives frames7/1/4 and shared D. For new routes North(A,B)/South(C,D), hold entire routes out rather than mixing route identity; two routes remain weak evidence.

  Separate pictured Cedar/Elm/Fir have3views each: train6views/2plants,test3/1; moving Fir view1 into training compromises unseen-plant evaluation.

- A model will be evaluated on new routes. Route North has trips A and B; route South has C and D. A proposed split trains on A and C and tests on B and D. Explain the problem and give a route-based alternative.

  Is this enough for a strong performance claim?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Assume the evaluation claim concerns photos of previously unseen plant identities, not additional views of already observed plants. The supplied task gives20 photos from5 plants but no exact frame assignments or model predictions. State a plant-level repair without inventing counts for a particular train/test allocation; keep all4views of each plant in one role.

  Explain the small-source-count limitation and do not claim a measured accuracy change.

- Each of five plants has four photos. A random photo split puts different views of every plant in both **training** and **test set**s. Explain the problem and repair.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Audit filenames, metadata, post-diagnosis fields, and duplicate plant identities; record every removed **leakage** path.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Distinguish frames from **group**s for Splits and related examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Ten photos per leaf are randomly split so one leaf appears in train and test.
- Explain and repair **leakage**.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A feature records the expert's final species decision, made after the **prediction time**. Why is this **leakage**, and what should happen to it?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 8 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

  Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

My reasoning and evidence

▼ Extension · Scaling without leakage — W1 Learn the scale from training

REFERENCE · SECTION 17

Extension · Scaling without leakage — W1 Learn the scale from training

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Using A,B,C, give the minimum, maximum and **range** for each feature. **Transform** all three **training** rows and query (11,120). Explain which rows supplied the scale.

> **My reasoning and evidence**

▼ Extension · Scaling without leakage — W2 Trace the changed prediction

REFERENCE · SECTION 18

Extension · Scaling without leakage — W2 Trace the changed prediction

> Retained extension task with its own values and teacher key.

Extension question and evidence

- For query (11,120), calculate raw and scaled distances to A,B,C. Give both k=1 **prediction**s. Does this prove that scaling always improves **prediction**s?

> **My reasoning and evidence**

▼ **Extension · Scaling without leakage — W3 Keep one coordinate system**

REFERENCE · SECTION 19

Extension · Scaling without leakage — W3 Keep one coordinate system

> Retained extension task with its own values and teacher key.

Extension question and evidence

- **Transform** (12,80) using the stored **training** scale. Explain the negative value and why validation must not fit its own scale. Give the worked validation and baseline scores.

My reasoning and evidence

▼ Extension · Scaling without leakage — W4 Scale a fresh dataset independently

REFERENCE · SECTION 20

Extension · Scaling without leakage — W4 Scale a fresh dataset independently

> Retained extension task with its own values and teacher key.

Extension question and evidence

- New **training** rows are P (20,1000) clear, Q (40,600) blocked, R (40,1000) blocked. Fit minima and **range**s. For query (22,680), show **transform**ed features, raw and scaled distances to P,Q,R, and both k=1 **prediction**s.

  Transform new row (50,520) with the same scale.

My reasoning and evidence

▼ Extension · Scaling without leakage — W5 Repair two preprocessing mistakes

REFERENCE · SECTION 21

Extension · Scaling without leakage — W5 Repair two preprocessing mistakes

> Retained extension task with its own values and teacher key.

Extension question and evidence

- A learner fits scales using **training** plus validation features but no labels. Another divides by a zero **training range**. Explain both problems and give a recorded repair for each.

> **My reasoning and evidence**

▼ Extra practice · Pictured case

REFERENCE · SECTION 23

Extra practice · Pictured case

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Split by plant identity when testing generalisation to new plants.
- Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/8

Split by plant, not by view

	view1	view2	view3
Cedar	train	train	train
Elm	train	train	train
Fir	test	test	test

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/8

Leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Group related views

All three Cedar views stay together; likewise Elm and Fir.

2. Assign groups

Training receives six views from Cedar/Elm; test receives three Fir views.

3. State claim

The final test probes one unseen plant, not three independent plants.

CHANGE ONE THING

Move Fir's first view into training while retaining its other two in test.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 24

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 25

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Split by plant identity when testing generalisation to new plants.
- Which labelled step matches this detail? The final test probes one unseen plant, not three independent plants.
- 1. Assign **group**s
- 2. State claim
- 3. **Group** related views
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: State claim The final test probes one unseen plant, not three independent plants. Read the picture in order.

The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Move Fir's first view into **training** while retaining its other two in test. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The unseen-plant evaluation is compromised.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The unseen-plant evaluation is compromised. Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Repair and keep the record

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

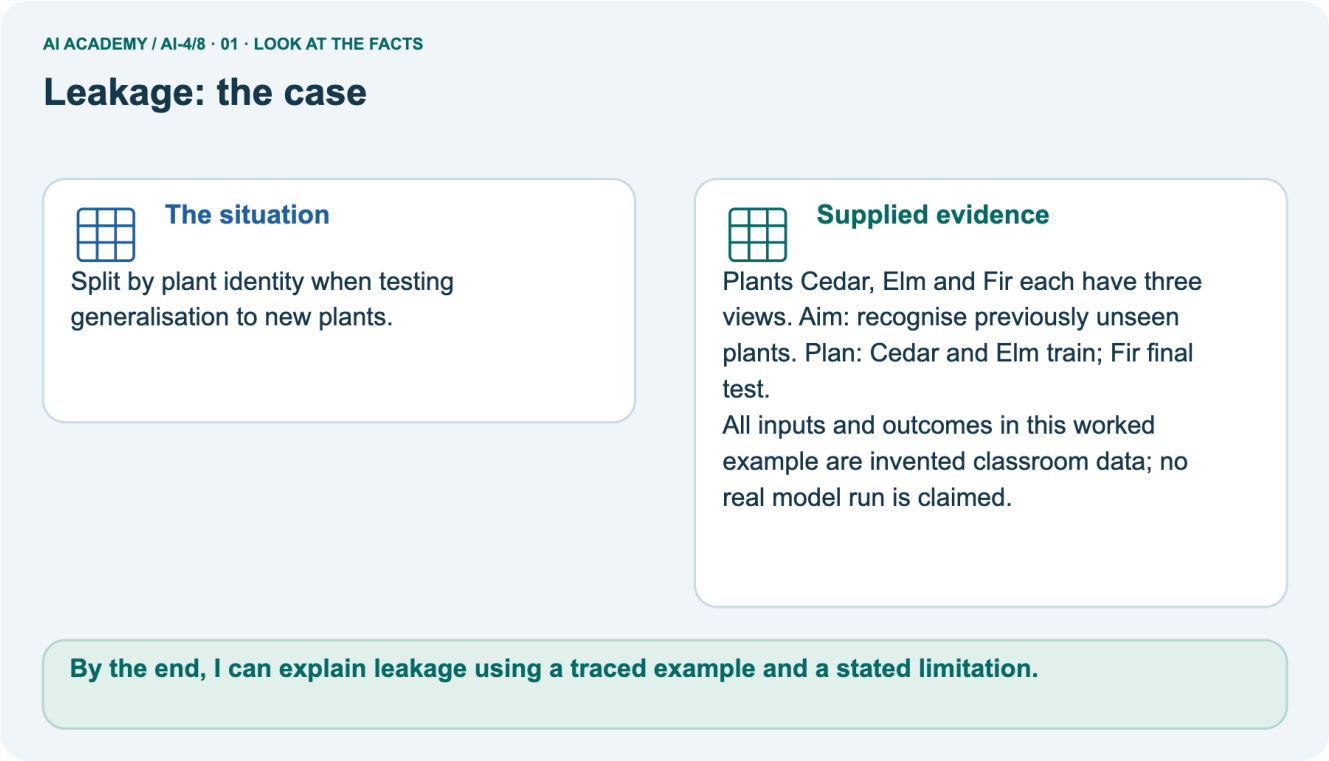
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Leakage: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

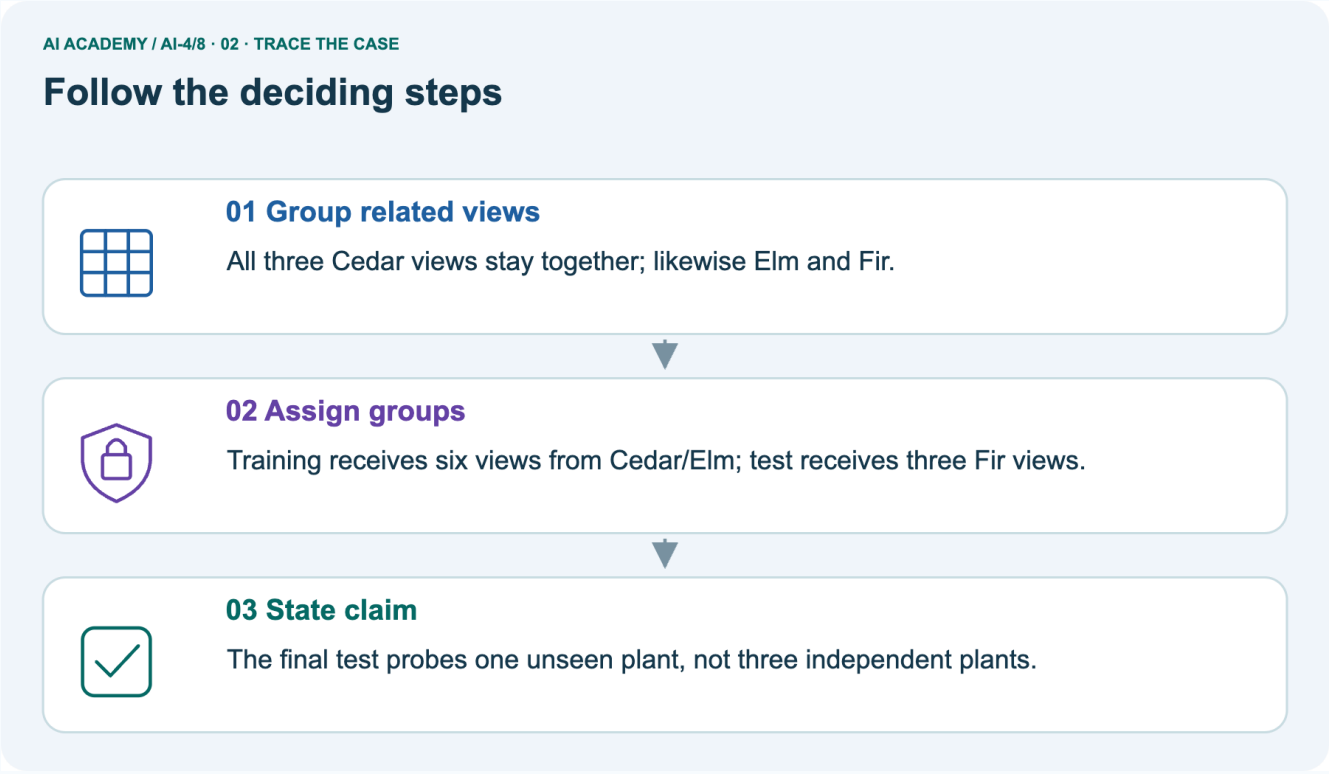
Split by plant identity when testing generalisation to new plants. Supplied facts: Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



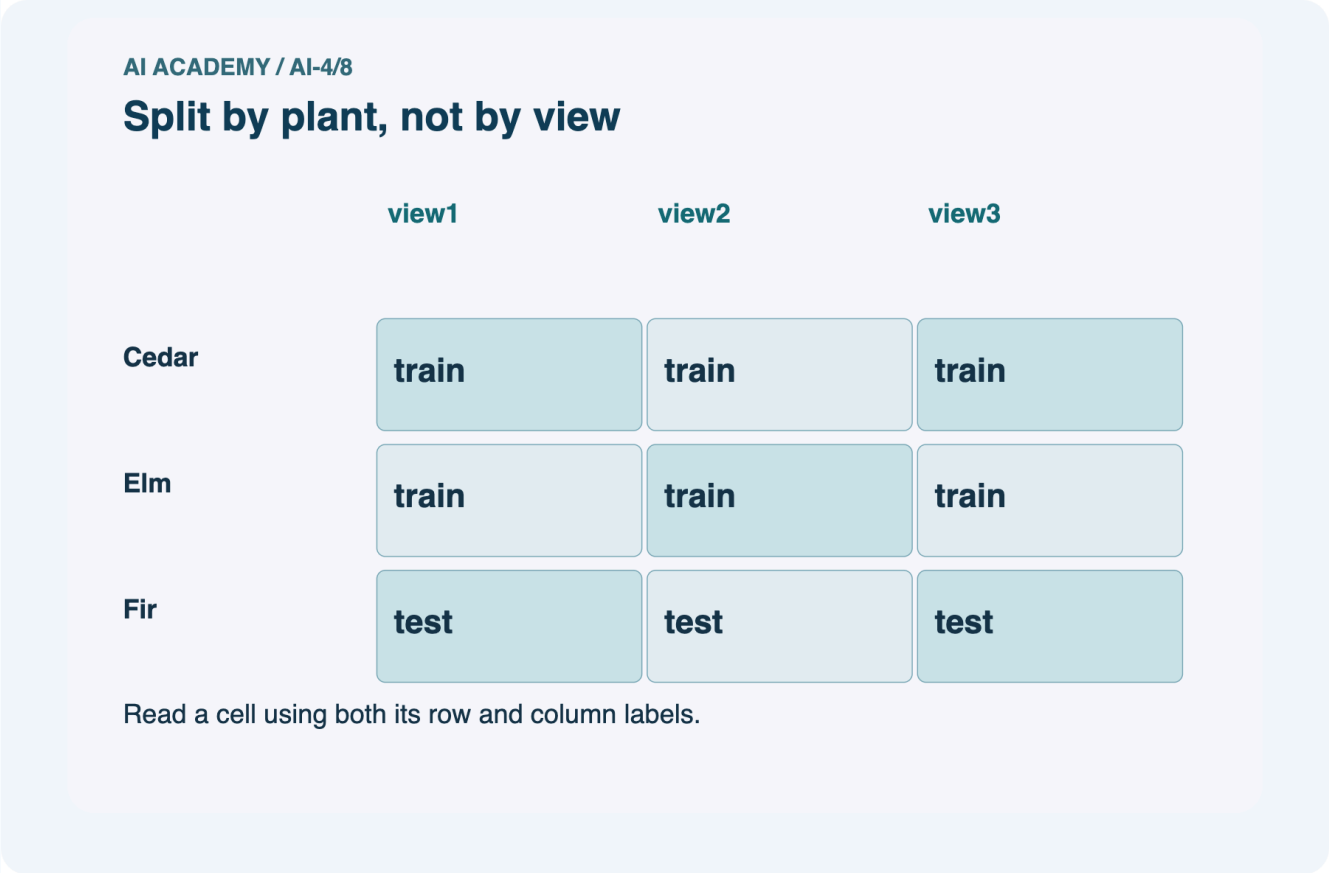
Follow the labelled picture with the matching explanation. Open the vector diagram

Group related views: All three Cedar views stay together; likewise Elm and Fir. → Assign **group**s: **Training** receives six views from Cedar/Elm; test receives three Fir views. → State claim: The final test probes one unseen plant, not three independent plants.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Split by plant, not by view



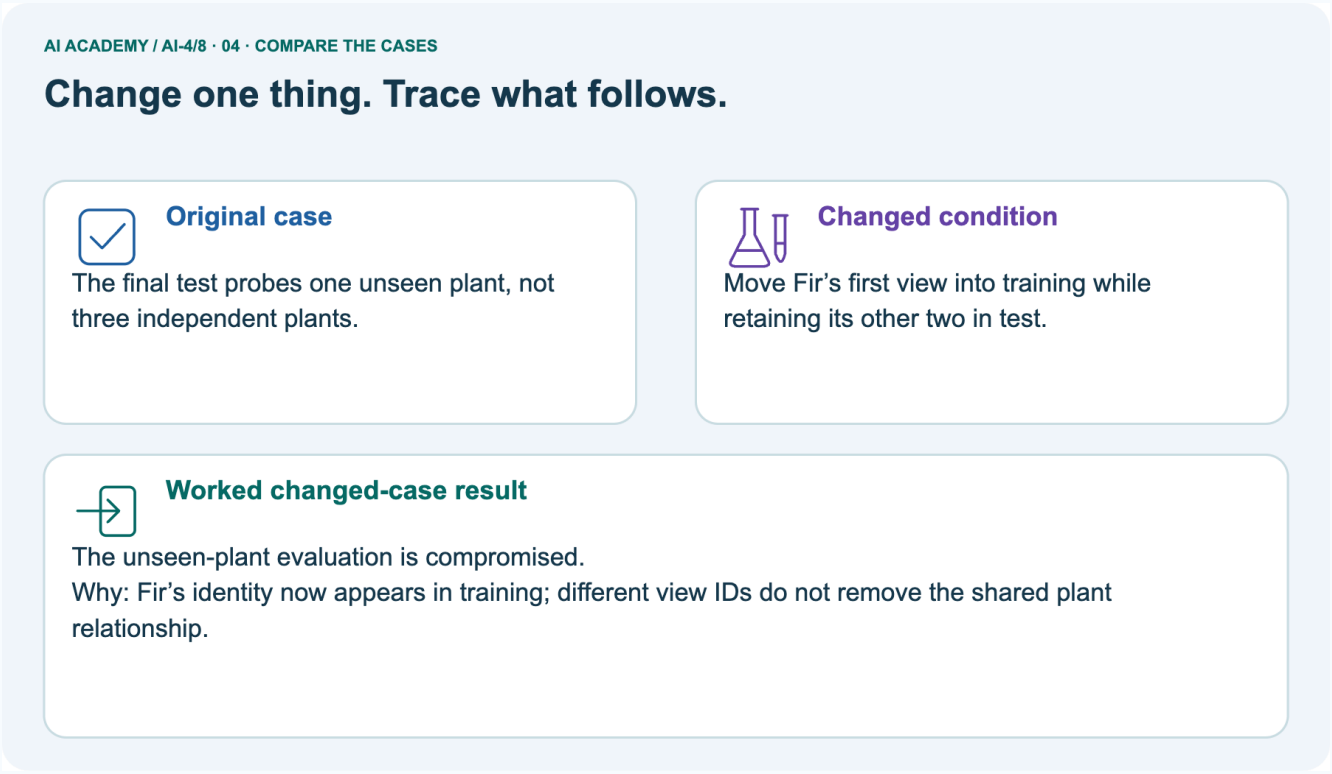
Follow the labelled picture with the matching explanation. Open the vector diagram

Split by plant, not by view. Cedar: view1 = train, view2 = train, view3 = train. Elm: view1 = train, view2 = train, view3 = train. Fir: view1 = test, view2 = test, view3 = test.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

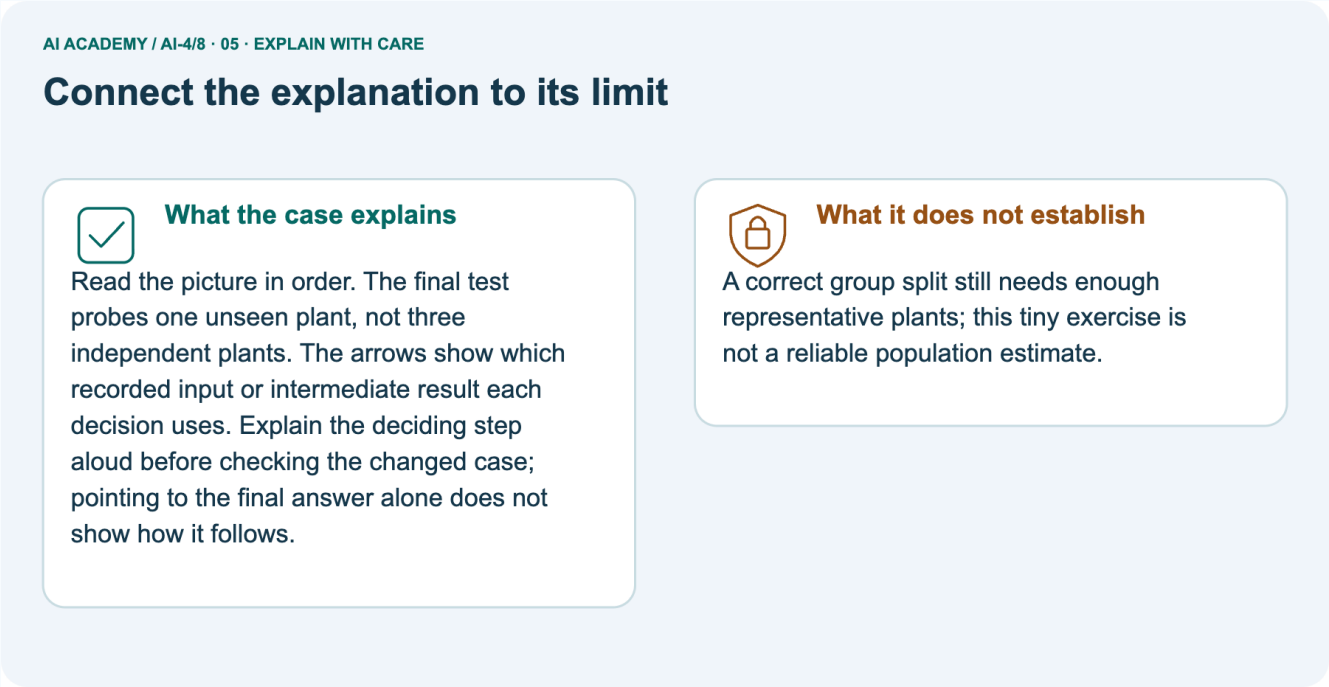
Original case: The final test probes one unseen plant, not three independent plants. Changed condition: Move Fir's first view into **training** while retaining its other two in test.

Expected result: The unseen-plant evaluation is compromised. Reason: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

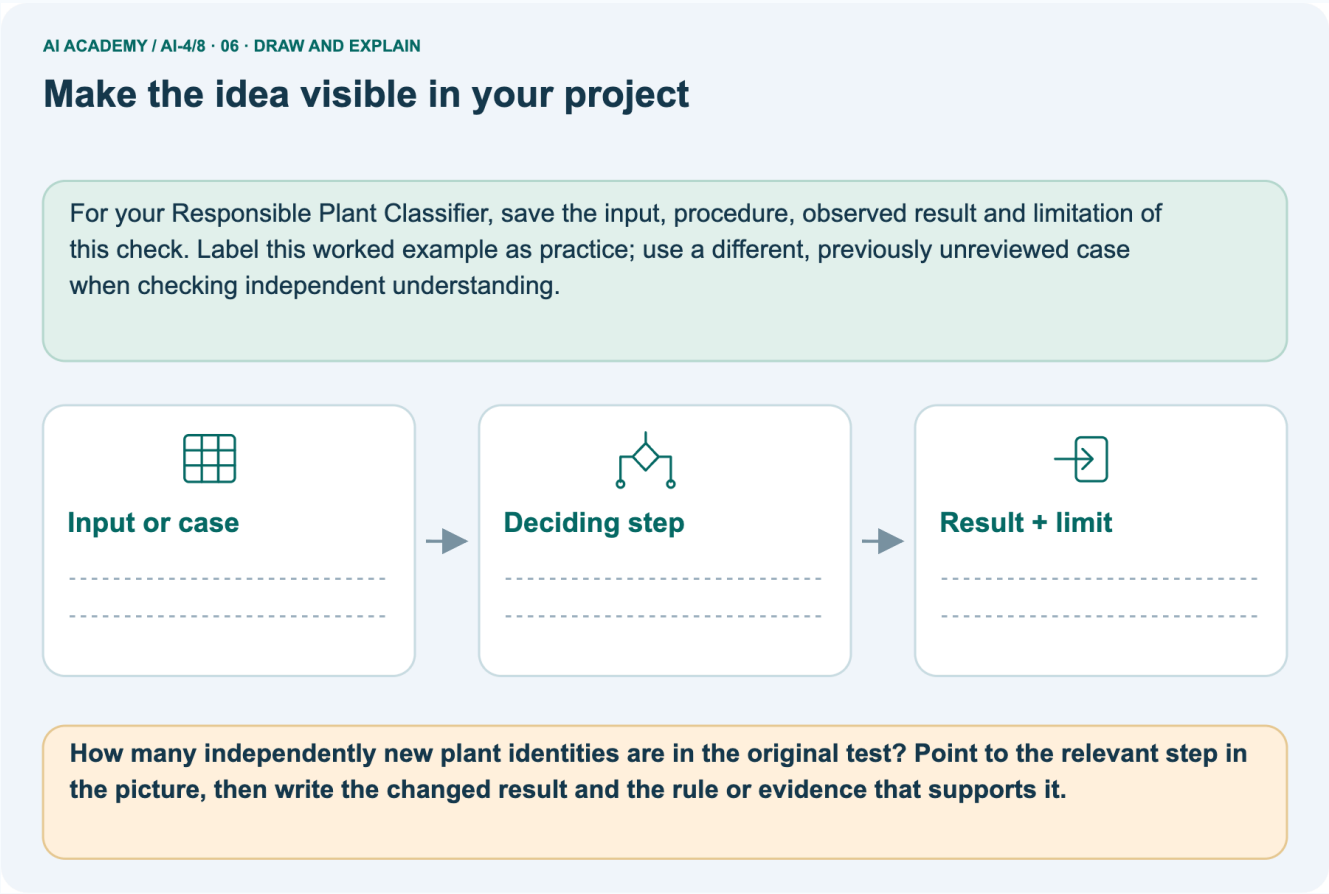
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correct **group** split still needs enough representative plants; this tiny exercise is not a reliable population estimate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Leakage: the case

02. Follow the deciding steps

03. Split by plant, not by view

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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